

The comparison interval is in bold and determined by the slowest run for each pair.

ALL EXPERIMENTS: **30 min timestep** and **ORCA025**

A. GOSI10.beta.5 + GOSI10.beta.4 SI3 namelist parameters : **1958-2000**

MLF (u-dy738) RK3 (u-dy739)

[Ocean validation notes on JASMIN](#)

[Sea ice validation notes on JASMIN](#)

B. GOSI10.beta.5 + GOSI10.beta.4 SI3 namelist parameters + GOSI10.beta.4 NEMO namelist parameters **1958-2000**

MLF (u-dy741) RK3 (u-dy742)

[Ocean validation notes on JASMIN](#)

[Sea ice validation notes on JASMIN](#)

C. GOSI10.beta.4 + GOSI10.beta.5 code changes up to (and including) the

[RK3 bug fix for penetrative solar radiation absorption over sea ice](#) (see the [NEMO issue](#))

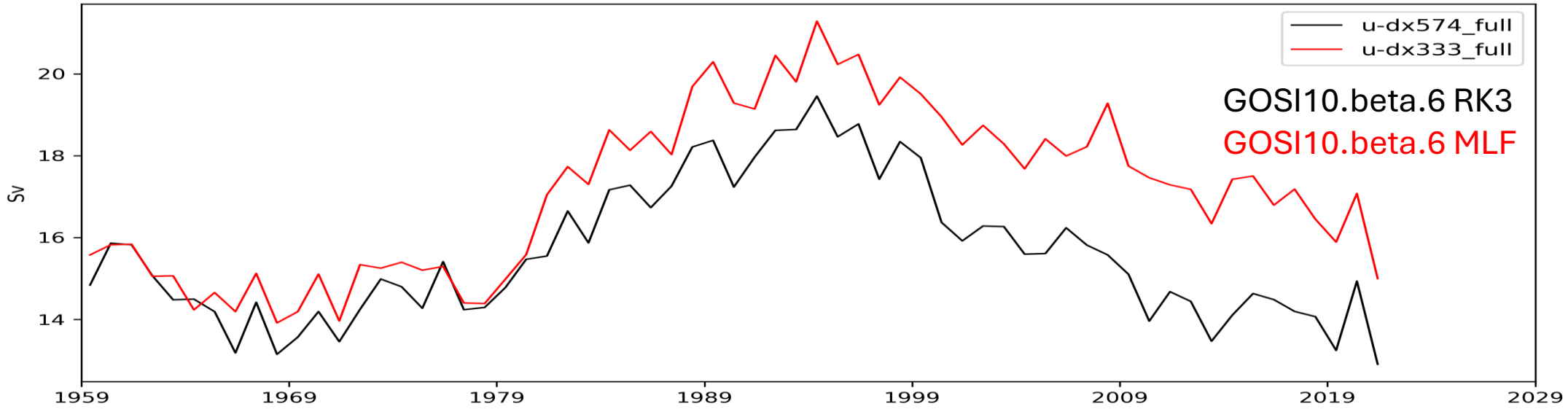
1958-1990 : this interval might not be long enough for a meaningful comparison

MLF (u-dy743) RK3 (u-dy746)

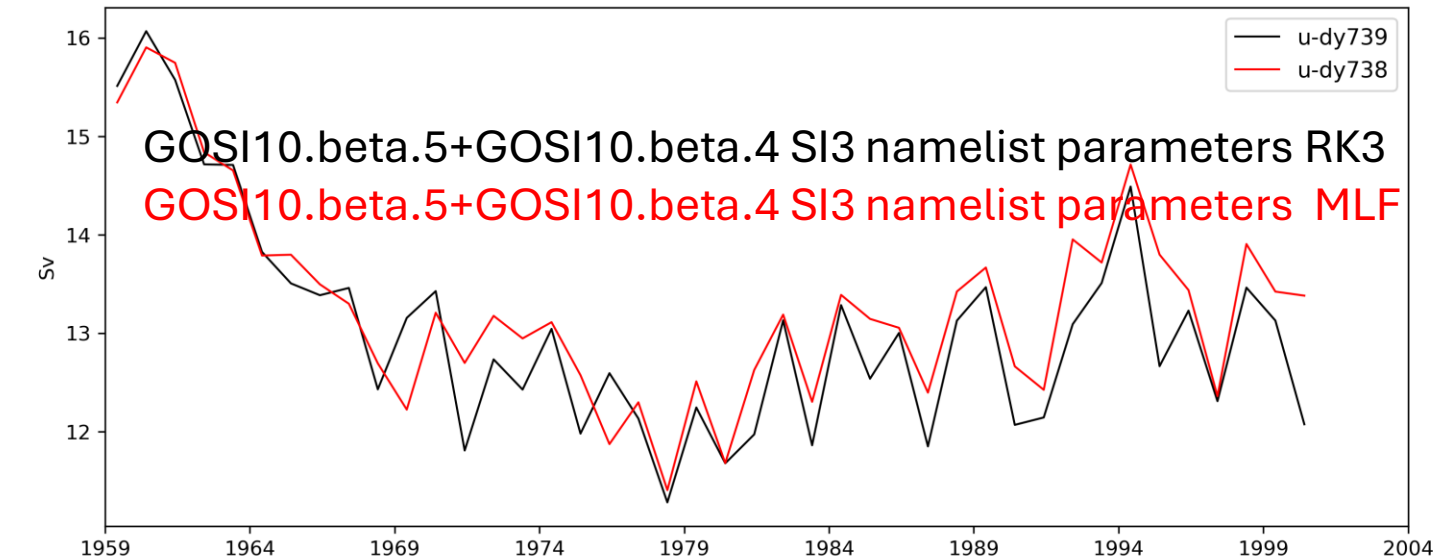
[Ocean validation on JASMIN](#)

[Sea ice validation notes on JASMIN](#)

Meridional overturning streamfunction depth-maximum (26N) (atlantic)



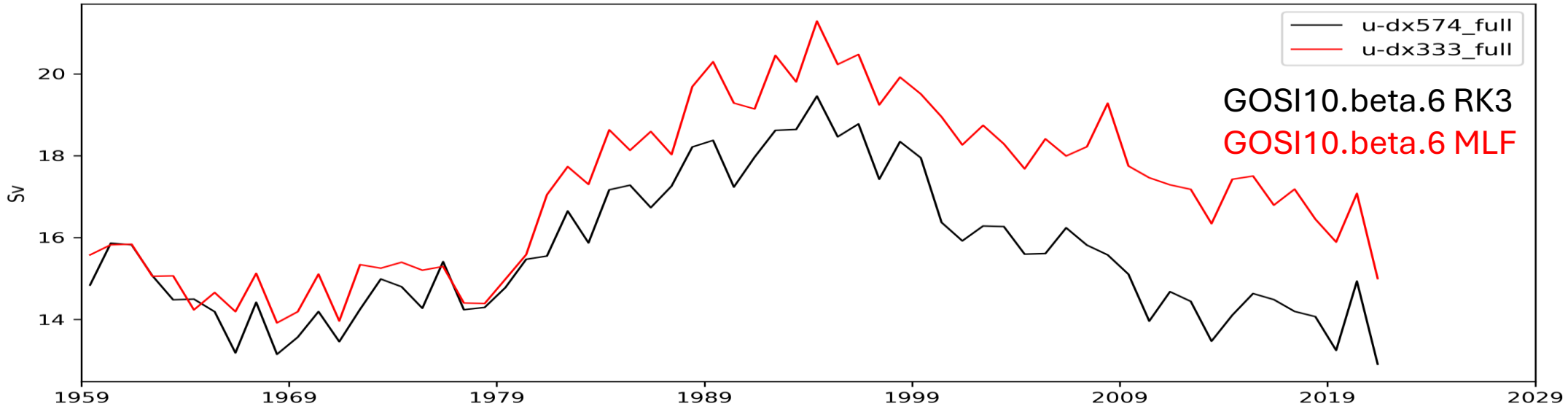
Meridional overturning streamfunction depth-maximum (26N) (atlantic)



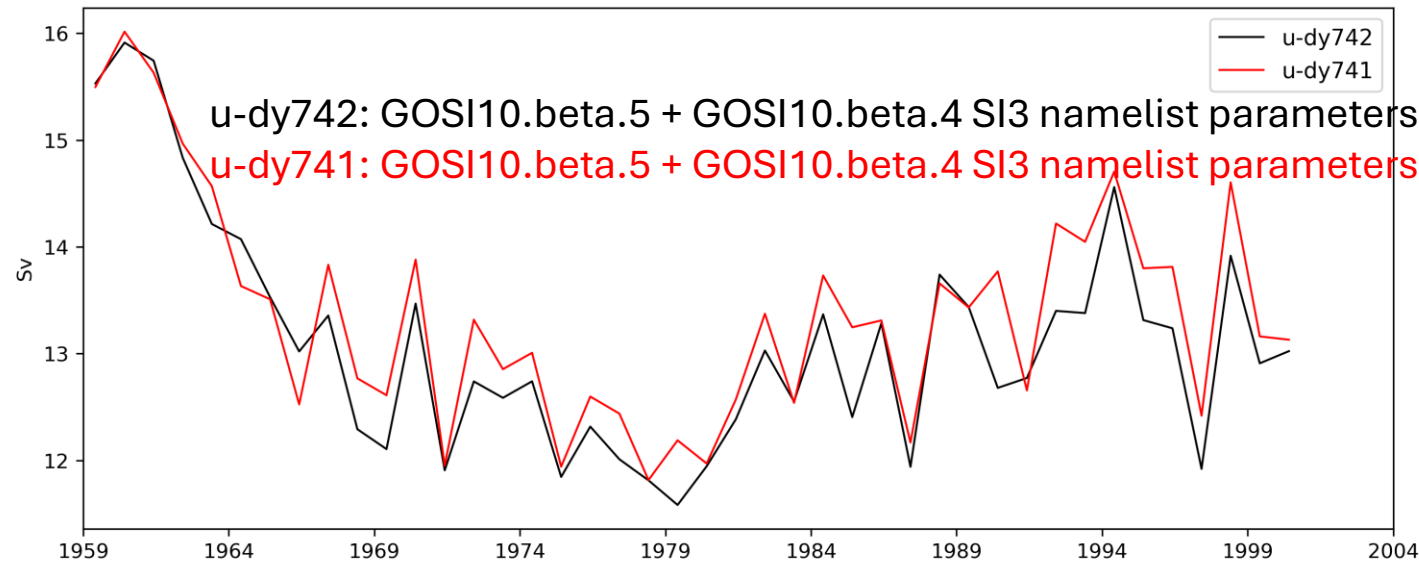
eORCA025 experiments **still**
running, this is just the
comparison so far

← So far experiments match closely, but
AMOC lower than GOSI10.beta.6 overall

Meridional overturning streamfunction depth-maximum (26N) (atlantic)



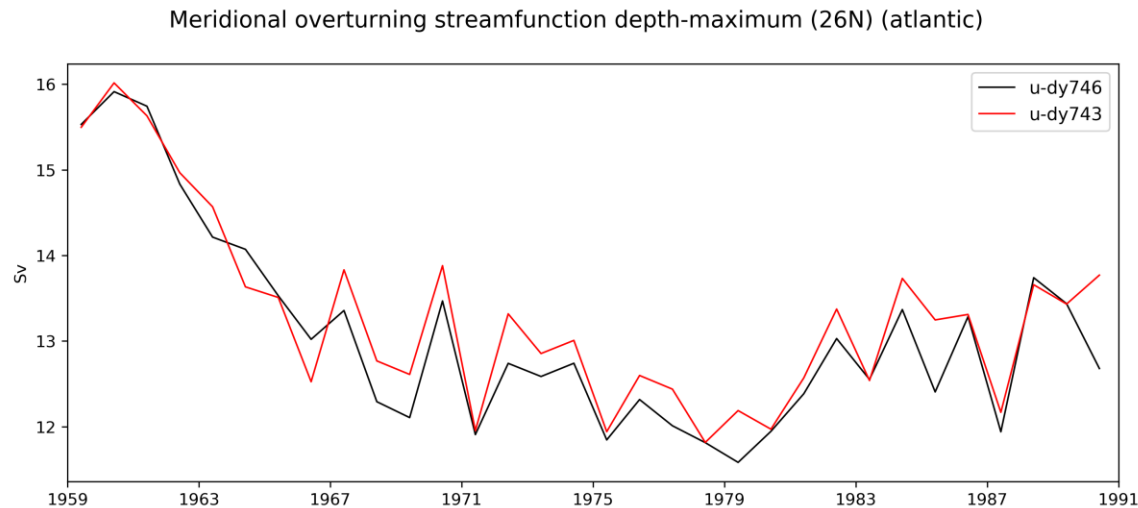
Meridional overturning streamfunction depth-maximum (26N) (atlantic)



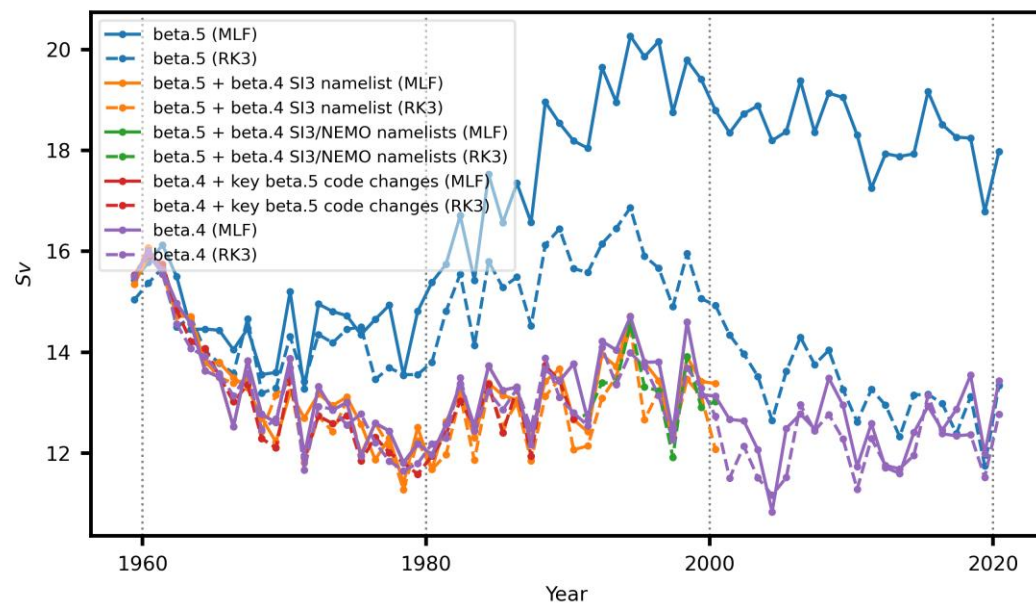
Similar to previous slide : eORCA025 experiments **still running**, this is just the comparison so far

← So far experiments match closely, but AMOC lower than GOSI10.beta.6 overall

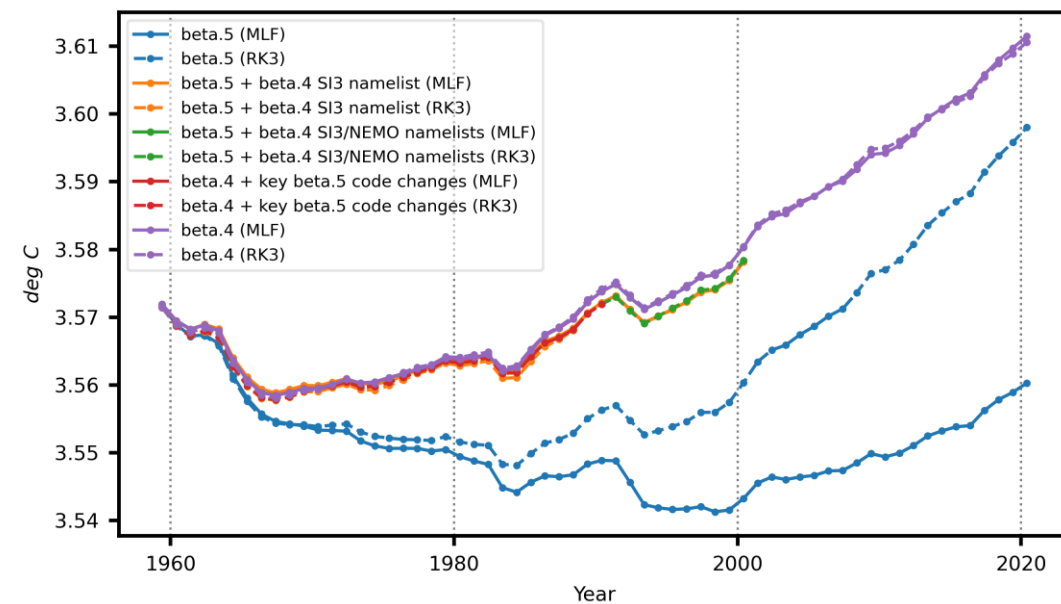
- **u-dy746: GOSI10.beta.4 + GOSI10.beta.5 code changes - RK3**
u-dy743: GOSI10.beta.4 + GOSI10.beta.5 code changes - MLF
- The comparison interval 1958-1990 might be too short so far (experiments still running)
- Similarly to the previous slides



Depth-maximum AMOC at 26N

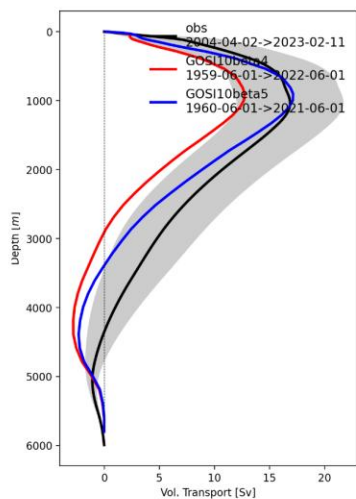
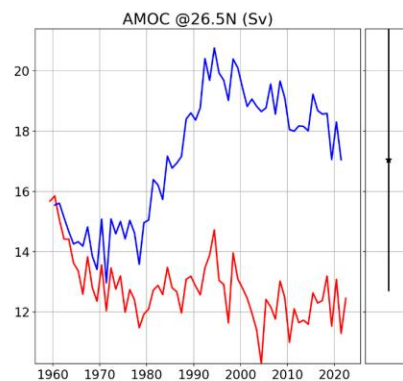
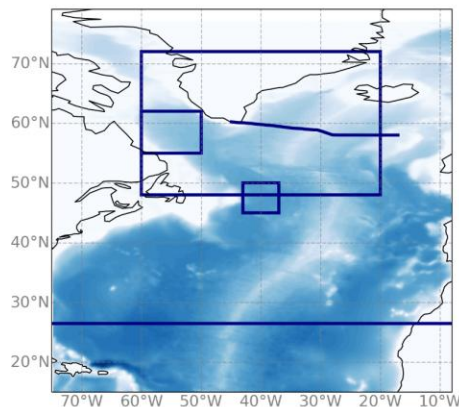
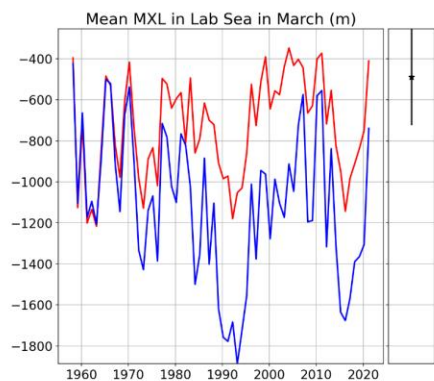
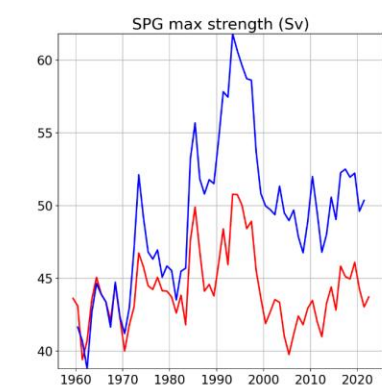


Global average conservative temperature



Solid lines: MLF
Dashed lines: RK3

GOSI10Releases



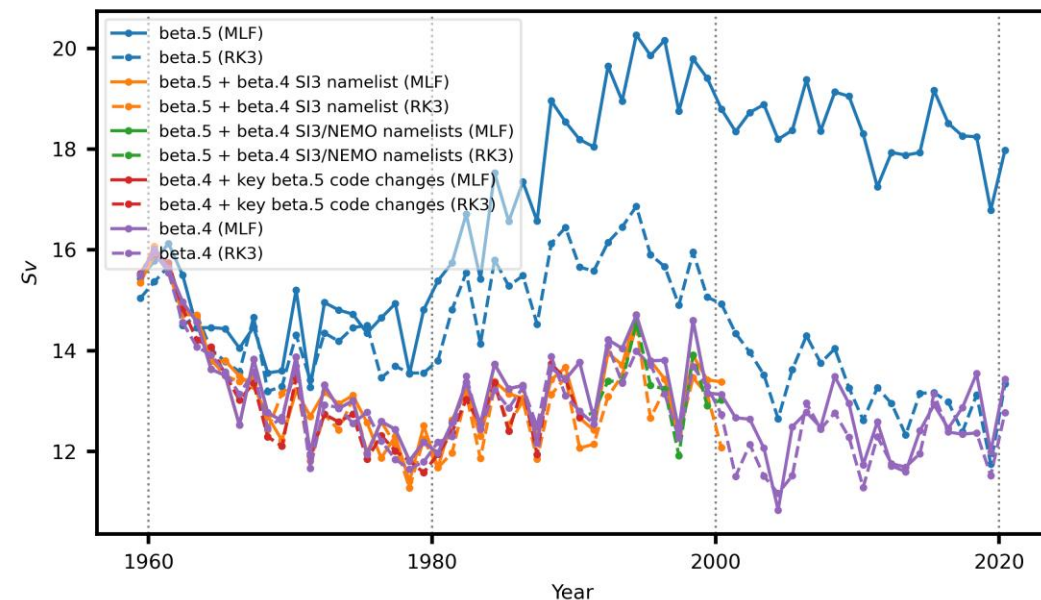
GOSI10.beta.5 MLF

GOSI10.beta.4 MLF

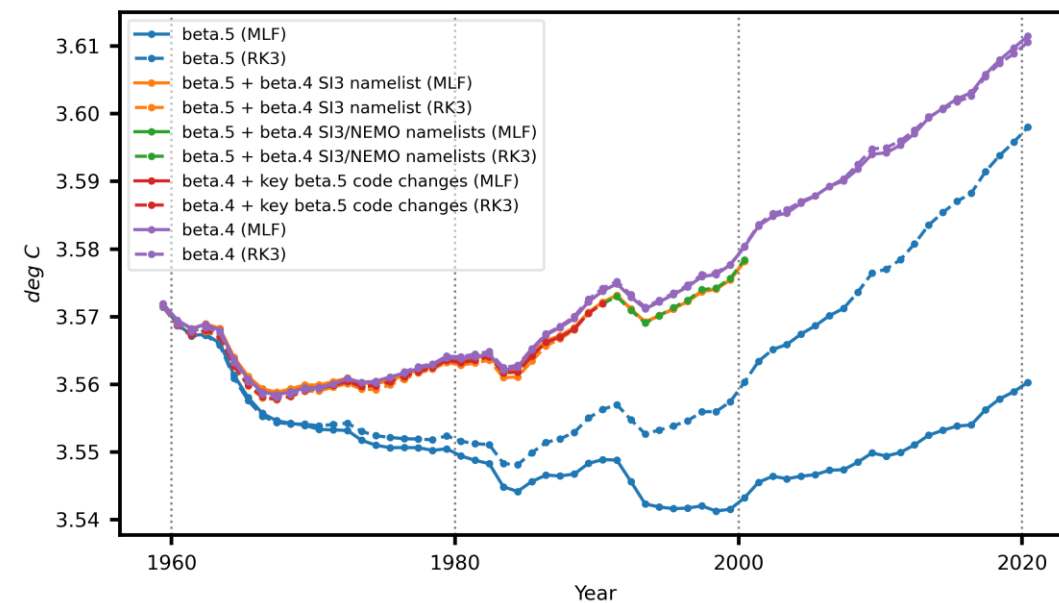
Solid lines: MLF

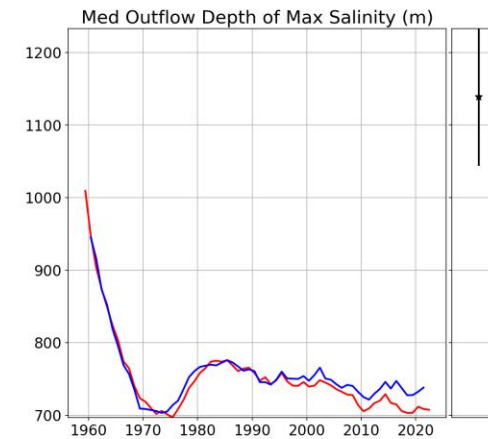
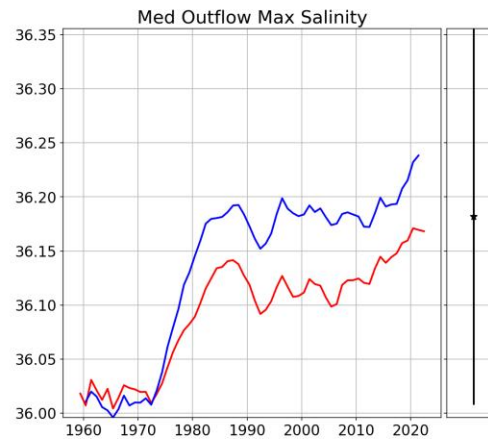
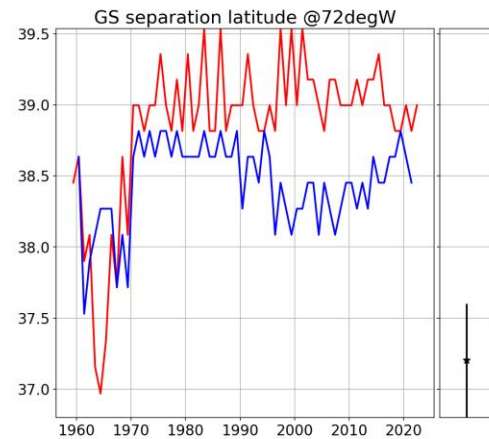
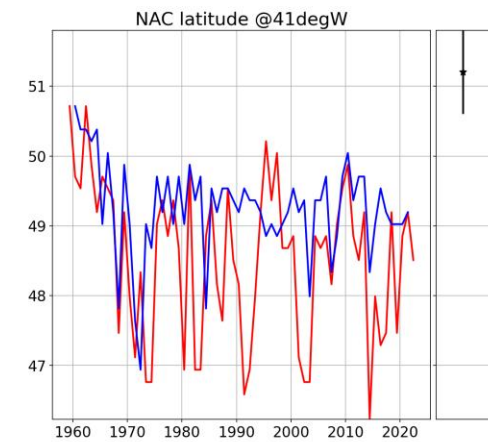
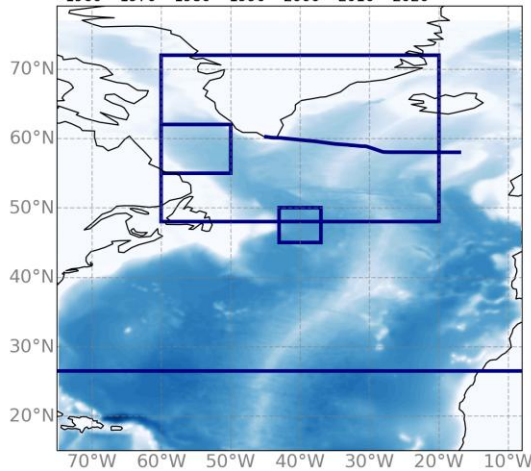
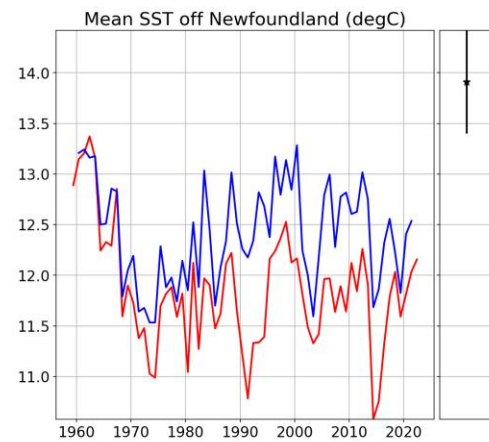
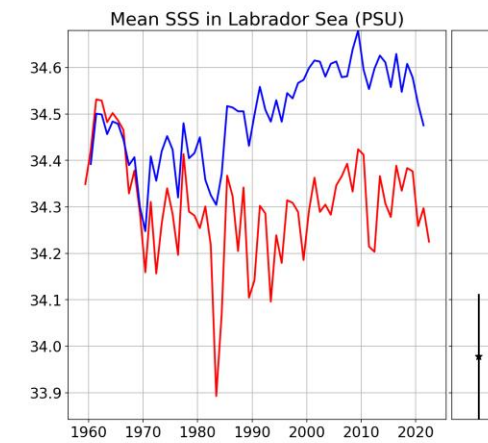
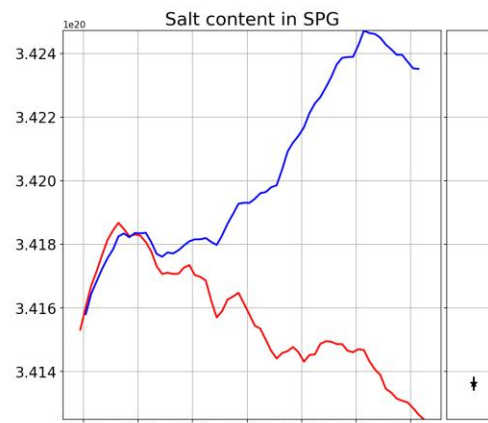
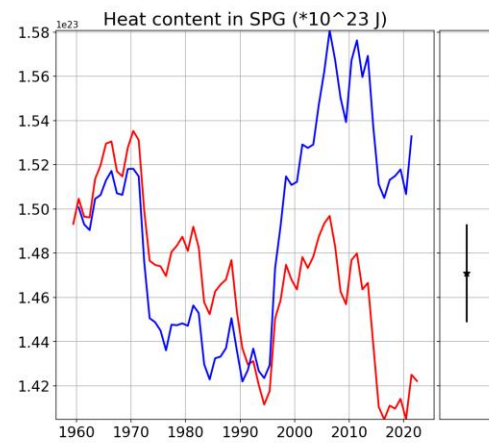
Dashed lines: RK3

Depth-maximum AMOC at 26N



Global average conservative temperature





— GOSI10beta4 — GOSI10beta5
 GOSI10beta4 = u-dq383 GOSI10beta5 = u-dw515

GOSI10Releases

GOSI10.beta.5 MLF
 GOSI10.beta.4 MLF

- **None of the latest sensitivity experiments exhibit the same MLF-RK3 discrepancies** as GOSI10.beta.6, so this is not triggered by any of the namelist changes introduced at GOSI10.beta.4
- It looks like it's the **SI3 namelist changes that trigger the abrupt increase in AMOC**, see validation notes from the GOSI10.beta.5 release where the increase in AMOC (to similar levels as in beta.6) coincides with a steep increase of SPG transport and salinification
- We now think that the differences RK3 vs MLF are due to RK3 (more stable) being able to somewhat mitigate the “dynamic instability/sensitivity” introduced by the SI3 changes, leading to a weaker AMOC wrt MLF